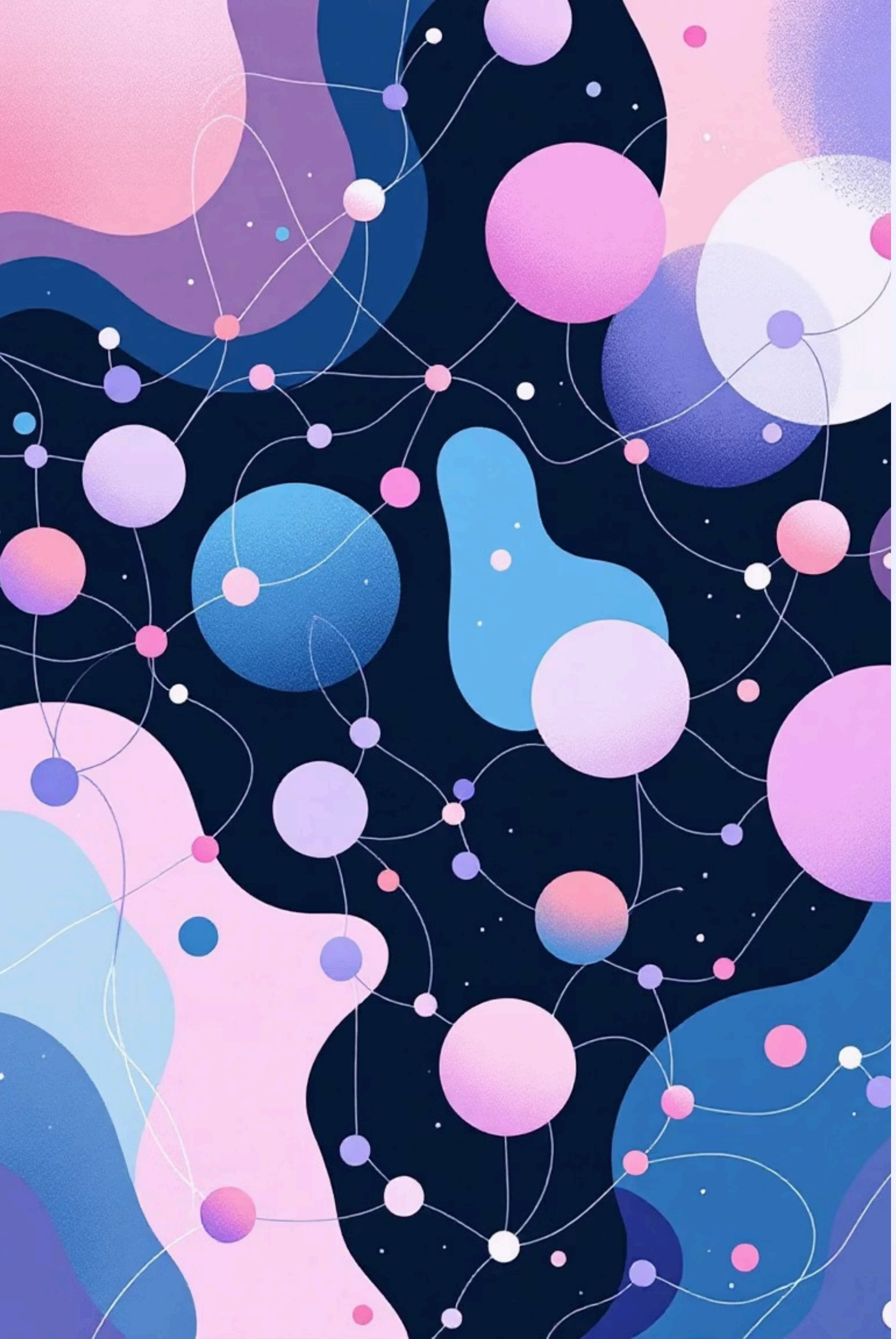


Types of Machine Learning

Understanding How Machines Learn from Data

Exploring the fundamental approaches that enable intelligent systems to learn, adapt, and make decisions.

What is Machine Learning?

Machine Learning is a subset of Artificial Intelligence where systems automatically learn patterns from data and make predictions or decisions without being explicitly programmed for every scenario.

The core principle: The methodology through which a system learns fundamentally determines its type and application domain.

Supervised Learning

Learning with guidance

Unsupervised Learning

Discovering hidden patterns

Semi-Supervised Learning

Best of both worlds

Reinforcement Learning

Learning through experience

Overview of ML Types

Each machine learning approach differs in how it processes data, what it aims to achieve, and where it excels in real-world applications.

Type	Data Used	Goal	Key Example
Supervised Learning	Labeled data	Predict outputs	Email spam detection systems
Unsupervised Learning	Unlabeled data	Find patterns	Customer segmentation for marketing
Semi-Supervised Learning	Partially labeled	Improve accuracy	Medical imaging diagnosis
Reinforcement Learning	Reward-based	Learn by experience	Autonomous vehicle navigation

Supervised Learning: Core Concept

What it is: Supervised learning utilises labeled datasets where both input features (X) and corresponding outputs (Y) are known. The model learns to map inputs to outputs by training on these correct examples.

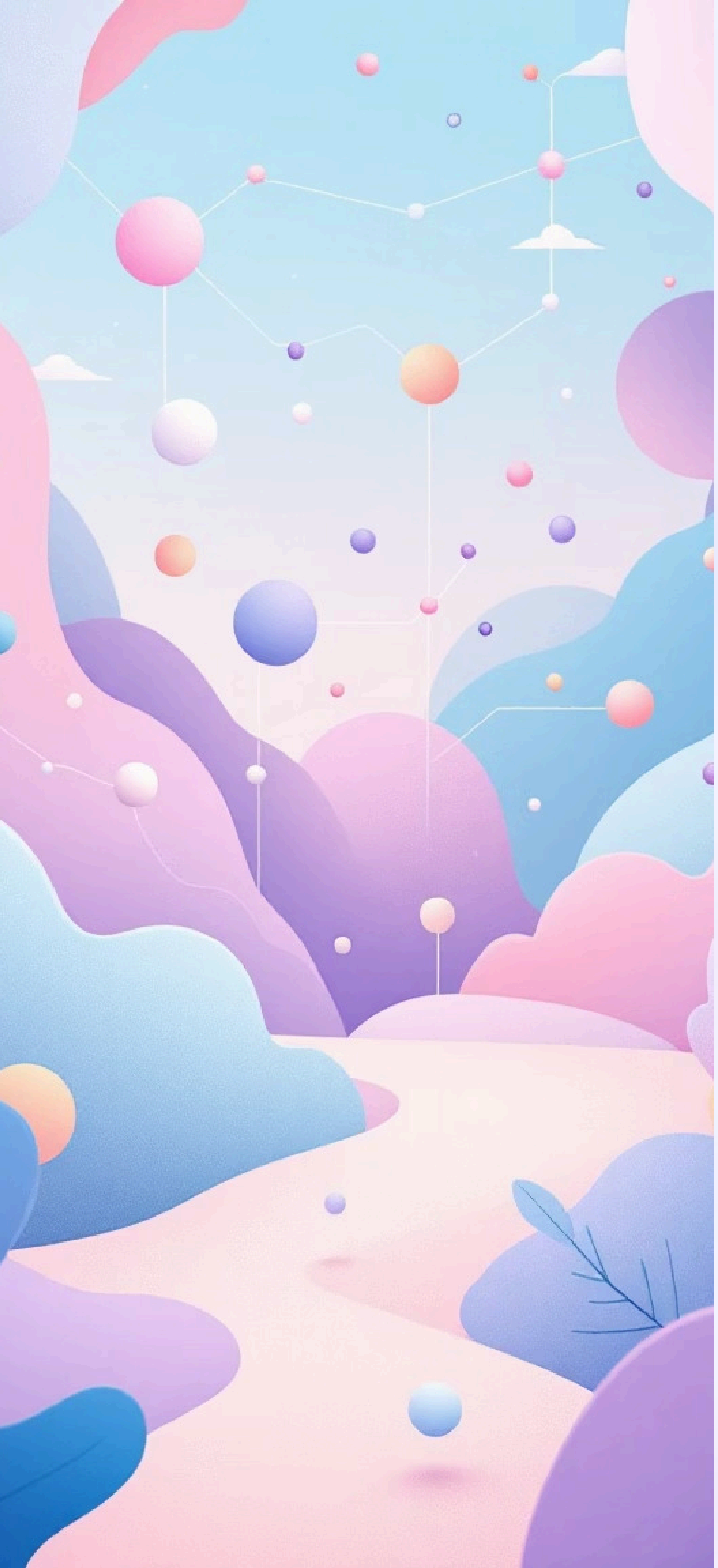
Training Process

During training, the algorithm analyses the relationship between features and labels, gradually improving its ability to predict outcomes for new, unseen data.

Example Dataset: Car Purchase Prediction

Age	Salary (₽)	Buy Car (Y)
25	₽40,000	No
30	₽70,000	Yes
35	₽60,000	Yes

Goal: Predict whether a person will purchase a car based on their age and salary characteristics.



Supervised Learning: Tasks & Algorithms

Regression Tasks

Purpose: Predict continuous numerical values

Example: House price prediction, temperature forecasting, stock market analysis

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor
- Support Vector Regressor (SVR)

Classification Tasks

Purpose: Predict discrete categorical outcomes

Example: Spam detection, disease diagnosis, image recognition

- Logistic Regression
- K-Nearest Neighbours (KNN)
- Decision Tree Classifier
- Random Forest Classifier
- Support Vector Machine (SVM)
- Naive Bayes

Supervised Learning: Evaluation & Applications

Performance Metrics

Task Type	Key Metrics
Regression	MAE, MSE, RMSE, R^2
Classification	Accuracy, Precision, Recall, F1-Score, ROC-AUC

These metrics help quantify model performance and guide optimisation efforts during development.

Property Valuation

Predicting residential and commercial property prices based on location, size, amenities, and market trends.

Email Filtering

Automatically detecting and filtering spam messages to protect users from unwanted content and phishing attempts.

Credit Assessment

Evaluating creditworthiness and loan eligibility by analysing financial history and risk factors.

Medical Diagnosis

Identifying diseases and health conditions from patient data, symptoms, and diagnostic test results.

Unsupervised Learning: Core Concept

What it is: Unsupervised learning works with unlabeled data — only input features (X) are provided, with no predetermined outputs (Y). The algorithm's objective is to discover hidden patterns, structures, or relationships within the data.

The model independently identifies groupings and associations without human guidance, making it ideal for exploratory data analysis.

Example: Income Clustering

Age	Annual Salary (₱)
22	₱25,000
25	₱28,000
45	₱85,000
48	₱90,000

Outcome: The model automatically clusters individuals into distinct groups such as "entry-level income" versus "senior professional income" based on natural patterns in the data.

Unsupervised Learning: Tasks & Algorithms

Clustering

Purpose: Group similar data points together based on feature similarity

Example: Customer segmentation for targeted marketing campaigns

Algorithms: K-Means, DBSCAN, Hierarchical Clustering, Gaussian Mixture Models

Association Rule Learning

Purpose: Discover interesting relationships between features in large datasets

Example: Market basket analysis revealing product purchase patterns

Algorithms: Apriori, FP-Growth, Eclat

Dimensionality Reduction

Purpose: Simplify complex data whilst preserving essential information

Example: Visualising high-dimensional data, feature extraction, data compression

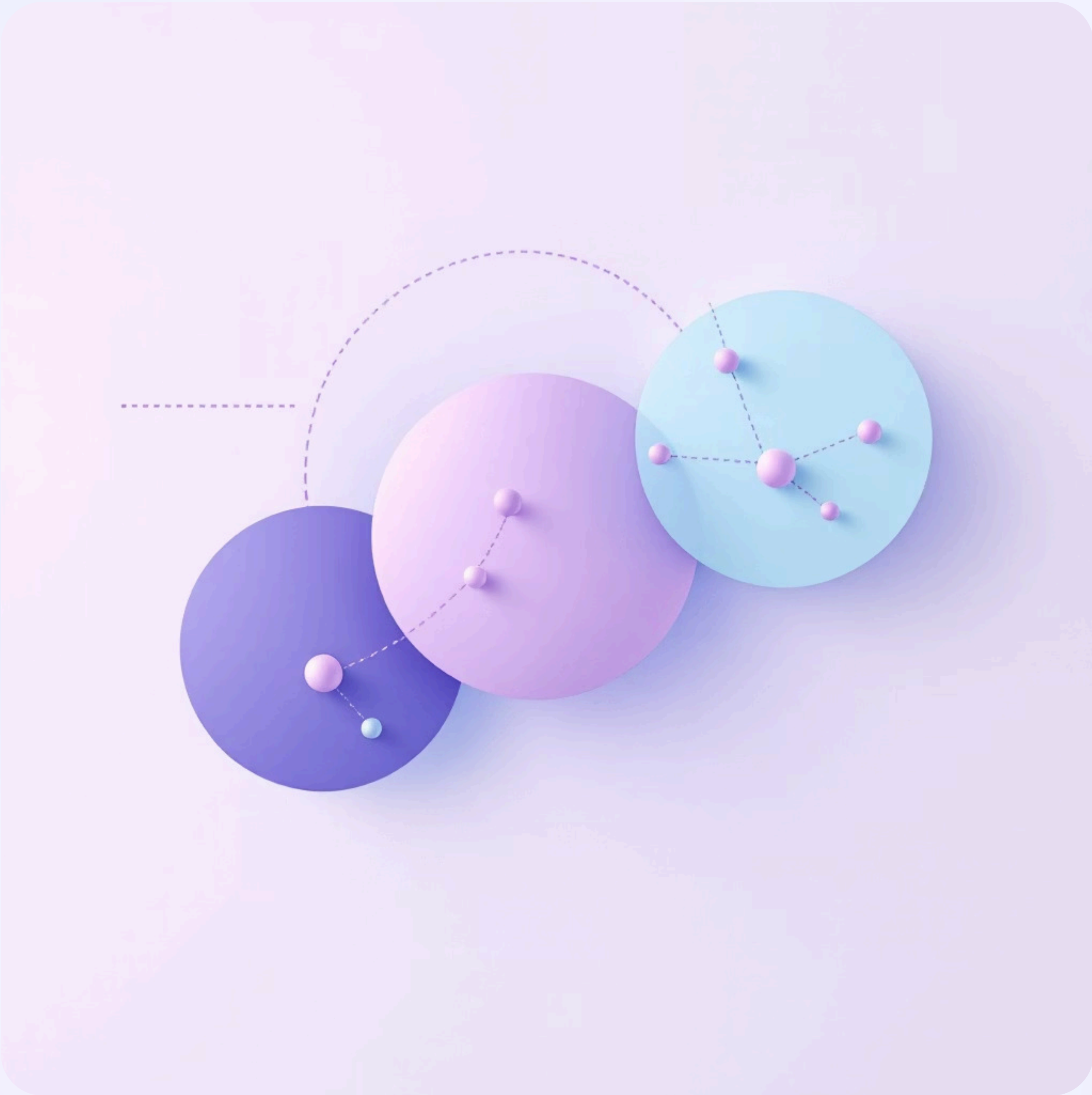
Algorithms: Principal Component Analysis (PCA), t-SNE, UMAP, Autoencoders

Unsupervised Learning: Evaluation & Applications

Evaluation Metrics

- **Silhouette Score:** Measures cluster cohesion and separation (ranges from -1 to 1)
- **Davies-Bouldin Index:** Evaluates average similarity between clusters (lower is better)
- **Inertia:** Within-cluster sum of squares (used in K-Means optimisation)

These metrics help assess clustering quality when ground truth labels are unavailable.



Market Basket Analysis

Uncovering product associations to optimise store layouts and create targeted promotions based on purchasing patterns.

Anomaly Detection

Identifying unusual patterns in financial transactions to detect fraud, system failures, or security breaches in real-time.

Customer Segmentation

Grouping customers based on behaviour, preferences, and demographics to deliver personalised experiences and targeted campaigns.

Summary: Supervised vs Unsupervised Learning

Type	Data Requirement	Primary Goal	Representative Example
Supervised Learning	Labeled datasets	Predict accurate outputs	House price prediction models
Unsupervised Learning	Unlabeled datasets	Discover hidden patterns	Customer segmentation analysis